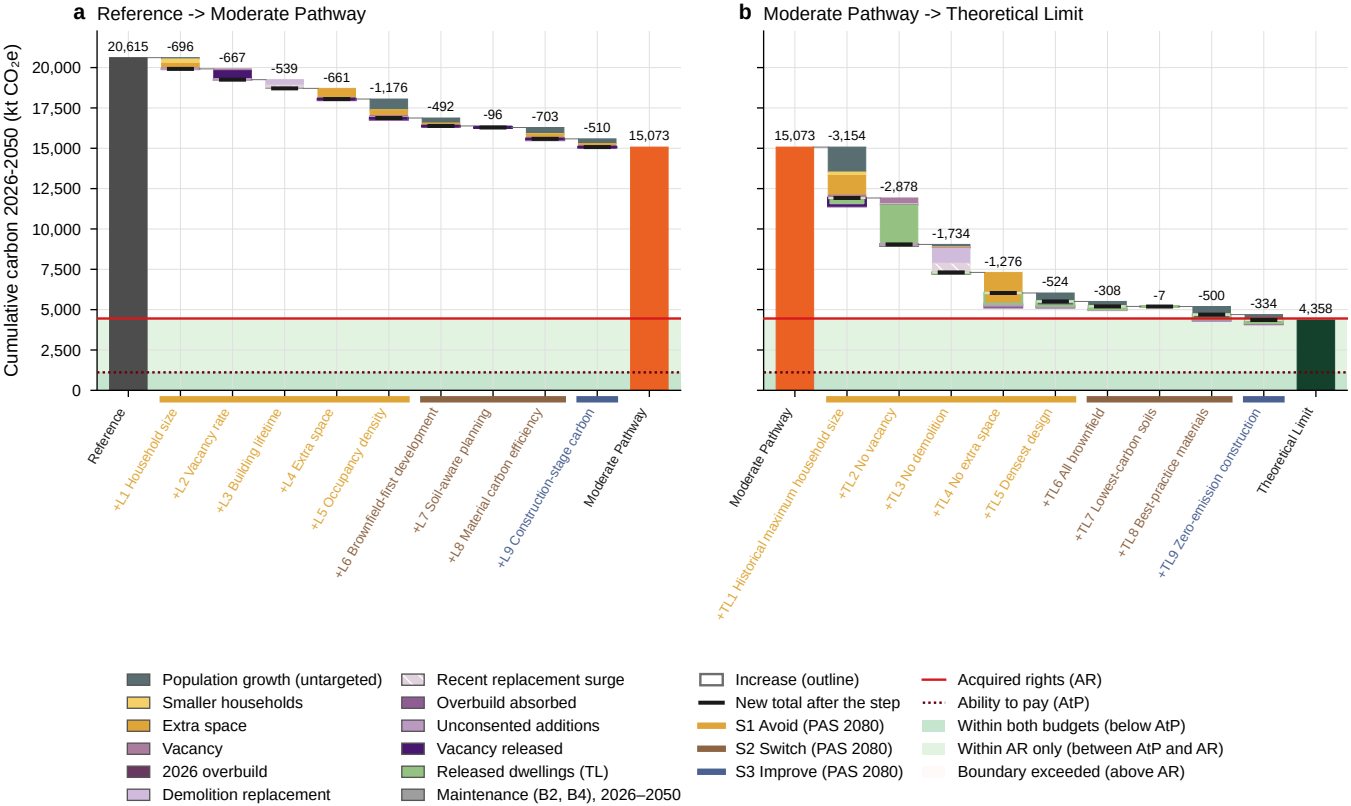
a Reference -> Moderate Pathway
b Moderate Pathway -> Theoretical Limit
Cumulative carbon 2026-2050 (kt CO2e)
20,000
17,500
15,000
12,500
10,000
7,500
5,000
2,500
0
20,615
-696
-667
-539
-661
-1,176
-492
-96
-703
-510
15,073
15,073
-3,154
-2,878
-1,734
-1,276
-524
-308
-7
-500
-334
4,358
Reference
+L1 Household size
+L2 Vacancy rate
+L3 Building lifetime
+L4 Extra space
+L5 Occupancy density
+L6 Brownfield-first development
+L7 Solrawate planning
+L8 Material carbon efficiency
+L9 Construction-stage carbon
Moderate Pathway
Moderate Pathway
+TL1 Historical maximum household size
+TL2 No vacancy
+TL3 No demolition
+TL4 No extra space
+TL5 Densest design
+TL6 All brownfield
+TL7 Lowest-carbon soils
+TL8 Best-practice materials
+TL9 Zero-emission construction
Theoretical Limit
Population growth (untargeted)
Smaller households
Extra space
Vacancy
2026 overbuild
Demolition replacement
Recent replacement surge
Overbuild absorbed
Unconsented additions
Vacancy released
Released dwellings (TL)
Maintenance (B2, B4), 2026–2050
Increase (outline)
New total after the step
S1 Avoid (PAS 2080)
S2 Switch (PAS 2080)
S3 Improve (PAS 2080)
Acquired rights (AR)
Ability to pay (AtP)
Within both budgets (below AtP)
Within AR only (between AtP and AR)
Boundary exceeded (above AR)